

SUMMARY

Applied scientist and research engineer with 6+ years of experience taking multimodal models from data design and pre/post-training through rigorous evaluation and production deployment. Led end-to-end VLM and visual-grounding projects, built scalable PyTorch training/serving systems, and contributed 40+ pull requests to Hugging Face Transformers.

EXPERIENCE

- **Toss Bank** Seoul, South Korea
Data Scientist Jan 2026 - Present
 - **On-premise AI Agent System:** Building an on-premise agent system with internally deployed LLMs for secure AI-assisted development workflows.
 - **AI Authentication:** Developing AI-based authentication systems for face and ID card verification.
 - **[Lead] End-to-End Document AI:** Led the full research and production pipeline for an approximately 1B-parameter vision-language model, spanning multi-page classification, orientation, layout detection, OCR/table understanding, and LLM-based key-value extraction; post-trained the model, designed stage-level diagnostics, and reached 61% exact-match accuracy for automation-ready outputs.
- **SuperbAI** Seoul, South Korea
Machine Learning Engineer Sep 2021 - Jan 2026
 - **[Lead] Multimodal Foundation Model:** Led multimodal pre-training and staged text/image alignment for a visual-grounding model; designed and curated 1.1M images with captions, boxes, and masks, then drove benchmark evaluation, API deployment, and AWS Marketplace release.
 - **CVPR 2025 Competitions:** Won 2nd place in IOD and 4th place in FSOD challenges, validating the model on open evaluation benchmarks.
 - **[Lead] ML Training & Serving Platform:** Built reusable distributed multi-GPU training, evaluation, and deployment infrastructure with PyTorch and AWS Batch; optimized production serving with TensorRT/Triton, improving throughput by 5x and delivering 1,100+ models and 60+ customer endpoints in one year.
 - **Post-training and Efficiency:** Applied LoRA, Adapter, and LST methods and memory-efficient training, reducing GPU memory by 65.6% and training time by 44.3% while preserving model performance.
 - **Interactive Segmentation:** Built browser, hybrid, and cloud-based segmentation tools using RepViT-SAM, FocalClick, SAM, and SAM2, increasing segmentation labeling speed by 25x.
- **Kakao Enterprise** Pangyo, South Korea
Machine Learning Engineer Intern Feb 2021 - Aug 2021
 - **AutoML:** Fixed AutoGluon NeuralNetFastAI scaling issue and developed a Flask-based training/inference AutoML framework with a simple front-end.
- **Team ISLAND** Seoul, South Korea
Chief Technology Officer Mar 2019 - Oct 2020
 - **Startup:** Raised approximately \$290K through government subsidies and seed funding; managed five developers across Android, Unity, and deep learning workstreams.
 - **Mobile ML:** Developed a lightweight 3D human pose estimation model for mobile devices and converted PyTorch models to TensorFlow Lite.

OPEN SOURCE

- **Hugging Face Transformers:** Contributed 40+ pull requests to a large production codebase across multimodal and vision models, independently shipping model architectures, processors, conversion scripts, distributed training fixes, tests, and documentation.
- **Hugging Face Transformers - SAM2:** Led the addition of Segment Anything 2 support to Transformers, including model integration, image/video processors, conversion flow, documentation, and tests; credited in the official [SAM2 docs](#).
- **Hugging Face Transformers - Molmo2:** Opened Molmo2 support for Transformers and published [danelesb/Molmo2-4B](#) on the Hugging Face Hub.

PAPERS

- **ZERO: Multi-modal Prompt-based Visual Grounding:** arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices:** IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array:** UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation:** Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease:** Sejong Science High School - Dec 31, 2013

EDUCATION

- **KAIST (Korea Advanced Institute of Science and Technology)** Daejeon, South Korea
Professional Master in Electrical Engineering; GPA: 3.55/4.3 Feb. 2020 – Aug. 2021
- **POSTECH (Pohang University of Science and Technology)** Pohang, South Korea
Bachelor in Electrical Engineering; GPA: 3.4/4.3 Mar. 2015 – Feb. 2020
- **UIUC (University of Illinois Urbana-Champaign)** Illinois, United States
Exchange Student; GPA: 3.6/4.0 Jan. 2018 – Dec. 2018

PROGRAMMING SKILLS

- **Research:** Python, PyTorch, Transformers, multimodal pre/post-training, LoRA, contrastive alignment, evaluation
Systems: Linux, AWS Batch, multi-GPU training, MLflow, TensorRT, Triton, ONNX, TensorFlow Lite, Git
Agents: LangChain, Langfuse, Claude Code, Codex, opencode (professional) **Languages:** Korean (native), English